

ER to Relational Model Mapping Rules

Step 1 : Entity

- From entity to relation/table
- Create a new relation that includes all the attributes
- Leave out multi-valued attributes (if any)
- Pick up appropriate PK

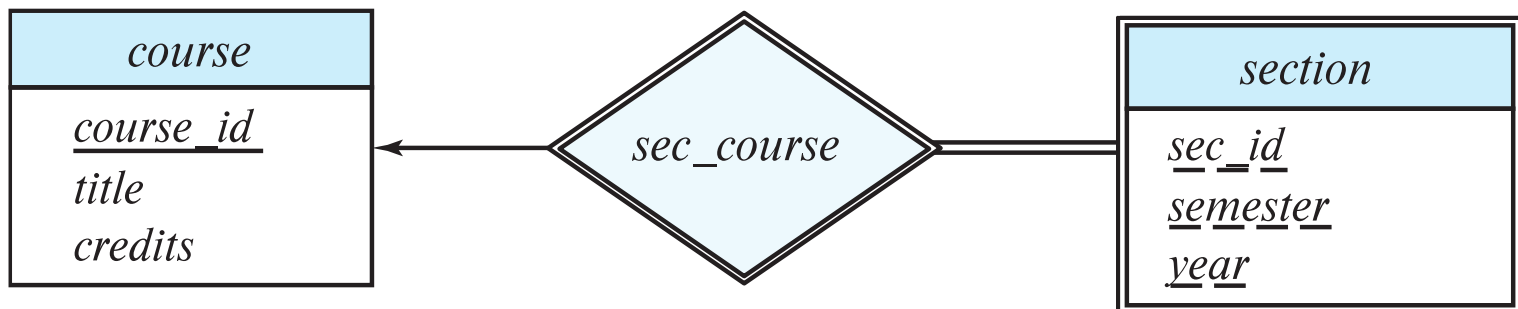
<i>instructor</i>
<u><i>ID</i></u>
<i>name</i>
<i>first_name</i>
<i>middle_initial</i>
<i>last_name</i>
<i>address</i>
<i>street</i>
<i>street_number</i>
<i>street_name</i>
<i>apt_number</i>
<i>city</i>
<i>state</i>
<i>zip</i>
{ <i>phone_number</i> }
<i>date_of_birth</i>
<i>age</i> ()

Instructor DDL

```
CREATE TABLE Instructor  
    (I_ID CHAR(20)  
    name CHAR(20),  
    dept_name CHAR(10),  
    salary (INTEGER)  
  
    PRIMARY KEY (I_ID) )
```

Step 2 : Weak Entity

- Create a new relation
- Include all attributes
- Include FK pointing towards owner relation
- Example: adding 'C_ID' to the 'section' relation
- PK is the combination of the FK and the partial key
- **Section**(C_ID, Sec ID, semester, year, *building*, *room_number*)



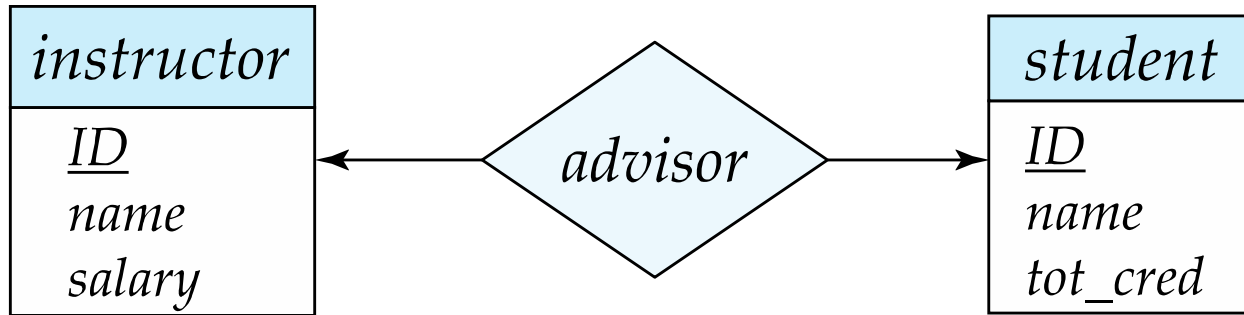
Weak Entity sets

- Weak entity set (section) and identifying relationship set (course) are translated into a single table Section_sec_course
- When the owner entity is deleted, all owned weak entities must also be deleted.

```
CREATE TABLE Section_Sec_course (  
  Sec_ID CHAR(20),  
  Semester CHAR (20),  
  Year CHAR (20),  
  Building CHAR(20),  
  Room number INTEGER,  
  C_ID INTERGER NOT NULL,  
  PRIMARY KEY (sec_ID, semester, year, C_ID),  
  FOREIGN KEY (C_ID) REFERENCES Course,  
  ON DELETE CASCADE)
```

Step 3 : 1:1 Relationship

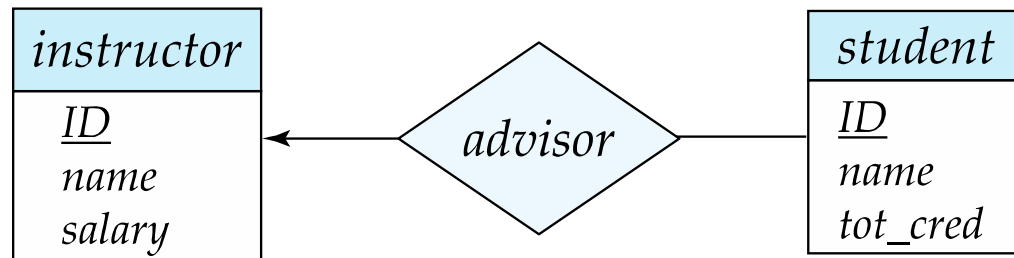
- Add to one of the participating relation a FK to other relation
- It is better to add to the relation that has a total participation in the relationship
- Include any relationship attributes
- Example: consider **instructor-advisor-student**
- Every student has at most one advisor and an instructor advises at most one student
- **Stu_advisor** (s_ID, s_name, dept_name, tot_credit, I_ID, since)
- Foreign Key I_ID references to **Instructor**



Every student must have an advisor

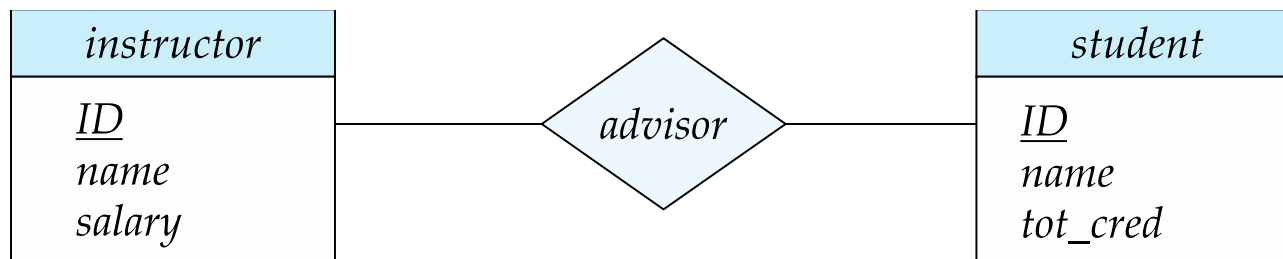
Step 4 : 1:N Relationship

- Add to the relation a FK from the other side
- Include any relationship attributes
- Example: instructor-advisor-student
- 1:N
- Add to student relation I_ID
- Add attribute since
- *Student (S_ID, S_name, dept_name, total_cred, I_ID, since)*
FK (I_ID) references Instructor



Step 5 : M:N Relationship

- Create a new relation containing FKs to both the participating relations
- Add relationship attributes (since)
- PK is the combination of both the FKs
- Instructor-advisor-student
- student can have many advisors and instructor can advise many students
- **Advisor(I_ID, S_ID, since)**



Step 6 : Multivalued Attributes

- Create a new relation
 - containing FK to the entity containing it
 - Attribute or attributes
- PK is FK plus the attributes
- Instructor (I_ID, Name, Dept_name, salary, {phone_no})
- *Inst_ph (I_ID, ph no)*
- It is all key relation
- FK (I_ID) references Instructor)

Step 7 : Ternary Relationship

- New relation containing a foreign key referencing each of the 3 entities
- Include relationship/descriptive attributes (if any)

Proj_guide (instructor, project, student)

- Proj_guide has descriptive attribute (since)
- **Proj_guide (I ID, P ID, S ID, since)**

